

Laboratory Equipment and Safety

Answer key with worked solutions

A. Hazards

1. Pictograms.

(a) **Flammable**: ethanol (b) **Oxidizer**: potassium nitrate or concentrated hydrogen peroxide (c) **Corrosive**: concentrated acids or sodium hydroxide (d) **Acutely toxic**: mercury compounds (e) **Toxic to aquatic life**: copper(II) sulfate. Other correct examples are acceptable.

2. Signal words and the SDS.

(a) **Danger** (Warning is less severe). (b) A **safety data sheet** gives a chemical's hazards, safe handling, storage, first aid and disposal. Read it **before** using the chemical, as part of the risk assessment.

3. Five unsafe actions.

1. Goggles pushed up: keep **goggles over the eyes** whenever chemicals are in use.
2. Water poured into concentrated acid: add the **acid slowly to water**.
3. Ethanol heated over a flame: it is flammable, so use a **water bath** or hot plate.
4. Test tube pointed at her partner: point it **away from everyone**.
5. Drinking in the lab: **no eating or drinking**. (Also: copper(II) sulfate is toxic to aquatic life, so it goes in the **waste container**, not down the sink.)

B. Choosing equipment

4. Equipment.

(a) **25 mL volumetric pipette** (b) **measuring cylinder** (or beaker) (c) **250 mL volumetric flask** (d) **burette** (e) **analytical balance**

5. Percent uncertainty.

$$(a) \frac{0.01 \text{ g}}{0.50 \text{ g}} \times 100\% = \mathbf{2\%} \text{ and } \frac{0.0001 \text{ g}}{0.5000 \text{ g}} \times 100\% = \mathbf{0.02\%}$$

$$(b) \frac{0.5 \text{ mL}}{10.0 \text{ mL}} \times 100\% = \mathbf{5\%} \text{ and } \frac{0.02 \text{ mL}}{10.00 \text{ mL}} \times 100\% = \mathbf{0.2\%}$$

The analytical balance and the pipette are about 100 and 25 times more precise. Small quantities need precise equipment, and a large cylinder is a poor choice for a small volume.

C. Burette readings

6. Titres.

(a) Rough: 24.10 mL – 0.00 mL = **24.10 mL**. Run 1: 24.70 mL – 1.20 mL = **23.50 mL**. Run 2: 23.85 mL – 0.40 mL = **23.45 mL**. Run 3: 23.55 mL – 0.05 mL = **23.50 mL**.

(b) Mean of runs 1–3:

$$\begin{aligned}\bar{V} &= \frac{(23.50 + 23.45 + 23.50) \text{ mL}}{3} \\ &= \frac{70.45 \text{ mL}}{3} = 23.483 \text{ mL}\end{aligned}$$

Deviations: +0.017 mL, –0.033 mL, +0.017 mL.

$$\begin{aligned}s &= \sqrt{\frac{1.67 \times 10^{-3} \text{ mL}^2}{3 - 1}} \\ &= 0.029 \text{ mL}\end{aligned}$$

Result: **23.48 ± 0.03 mL (n = 3)**. The rough titre is left out: it only shows roughly where the end point is.

7. The meniscus.

Water forms a curved surface that dips in the middle; the calibration marks are set to the **bottom** of the curve. With the eye above or below, the meniscus lines up with the wrong mark: this is **parallax error**.

D. Challenge

8. Titre uncertainty.

(a) Two readings: $2 \times 0.05 \text{ mL} = 0.10 \text{ mL}$.

$$\frac{0.10 \text{ mL}}{23.50 \text{ mL}} \times 100\% = \mathbf{0.43\%}$$

(b) Make the titre **larger**, for example by using a more dilute titrant or a larger volume of the solution being titrated, so the same $\pm 0.10 \text{ mL}$ is a smaller fraction of it.